DATA ANALYSIS COURSE | CAPSTONE PROJECT

HR Workforce & Attrition Analysis

Project brief and key analysis questions for student teams

1. Project Overview

You are a data analyst on the People Analytics team of a mid-sized company with a headquarters in Ohio and a growing remote workforce. Leadership wants to understand who works at the company, how hiring has evolved, and why people leave. Your job is to clean the HR dataset, answer the key questions below, and present clear, evidence-based recommendations.

Business objectives

- Understand the current workforce: size, structure, and diversity.
- Measure hiring and termination trends over time.
- Identify which departments, roles, and locations have the highest turnover.
- Recommend actions to improve retention and workforce balance.

2. Dataset at a Glance

File: HR_Data.csv (one row per employee, current and former).

Item	Detail
Size	22,214 employees x 13 columns; no duplicate employee IDs
Identity	id, first_name, last_name (personal data; do not publish names in your final deliverable)
Demographics	birthdate, gender (Male, Female, Non-Conforming), race (7 categories)
Job	department (13), jobtitle (185 distinct titles)
Location	location (Headquarters / Remote), location_city (77), location_state (7 states)
Dates	hire_date, termdate (blank = still employed)
Employment status	3,929 employees have a termination date (about 17.7%); 18,285 are blank (active)

3. Data Quality Challenges (Start Here)

Real HR data is messy. Part of your grade is how well you detect, document, and fix these issues before analysis.

Issue found	What you need to do
Mixed date formats	birthdate and hire_date appear in at least three styles (e.g. 1/20/2002, 04-08-19, 06/04/1991). Standardize to one format and document your rules.
Two-digit years	Values like 04-08-19 are ambiguous (2019 or 1919?). Decide a rule, justify it, and check that no one is hired before age 16 or born in the future.
Termination timestamps	termdate is a full UTC timestamp (e.g. 2023-11-23 21:27:59 UTC). Convert to a date and create an Active / Terminated status column.
Future termination dates	About 1,000 termination dates fall after today (some as late as 2041). Decide how to treat them: scheduled exits, errors, or exclude. State your assumption clearly.
Logical checks	Verify termdate is not earlier than hire_date, and that location (Remote) is consistent with city and state.
Derived fields	Create age, age group, tenure (years), hire year, termination year, and an is_terminated flag.

4. Key Analysis Questions

Answer every question with a table or chart and one sentence of business interpretation. Questions are grouped by theme.

A. Workforce Composition

Q1. How many employees does the company have in total, and how many are currently active vs. terminated?
Hint: Use the is_terminated flag you created.

Q2. What is the headcount by department, and which departments are the largest and smallest?

Q3. What is the gender distribution overall, and how does it differ across departments?

Q4. What is the age distribution of the workforce, and which age group is the largest?
Hint: Calculate age as of a fixed reference date.

Q5. What are the most common job titles, and how are titles spread across departments?

B. Diversity & Inclusion

Q6. What is the racial and ethnic composition of the company overall and by department?

Q7. Is there a gender or race imbalance in specific departments or job titles?

Q8. Do gender and race representation differ between Headquarters and Remote employees?

Q9. Are turnover rates different across gender and race groups? Report rates, not just counts.
Hint: Compare group size before drawing conclusions.

C. Hiring Trends

Q10. How many employees were hired each year, and what is the overall hiring trend?

Q11. Which years had the highest and lowest hiring, and what might explain them?

Q12. How has hiring by department changed over time?

Q13. Is the share of remote hires growing compared with headquarters hires?

D. Attrition & Turnover

Q14. What is the overall termination (turnover) rate?

Q15. How many employees left each year, and is turnover increasing or decreasing?
Hint: Handle the future-dated terminations according to your documented rule.

Q16. Which departments have the highest and lowest turnover rate (not just count)?

Q17. Which job titles show the highest turnover, and do they cluster in particular departments?

Q18. Do remote employees leave at a different rate than headquarters employees?

Q19. How does turnover vary by age group and by gender?

E. Tenure & Retention

Q20. What is the average tenure of employees who left vs. those still active?

Q21. In which year of employment do people most often leave (first year, 2-3 years, 5+ years)?

Q22. Which departments retain employees the longest and which lose them the fastest?

Q23. Is there a relationship between age at hire and tenure?

F. Geography & Work Location

Q24. What share of the workforce is at Headquarters vs. Remote, and which departments rely most on remote work?

Q25. Which states and cities have the most employees, and how concentrated is the workforce in Ohio?

Q26. Do turnover rates differ by state or city?

G. Strategic Synthesis

Q27. Which three departments or segments are at the greatest retention risk, and what evidence supports this?

Q28. What are your top five data-driven recommendations for HR leadership?

Q29. What additional data (e.g. salary, performance, exit reasons, manager) would make this analysis stronger?